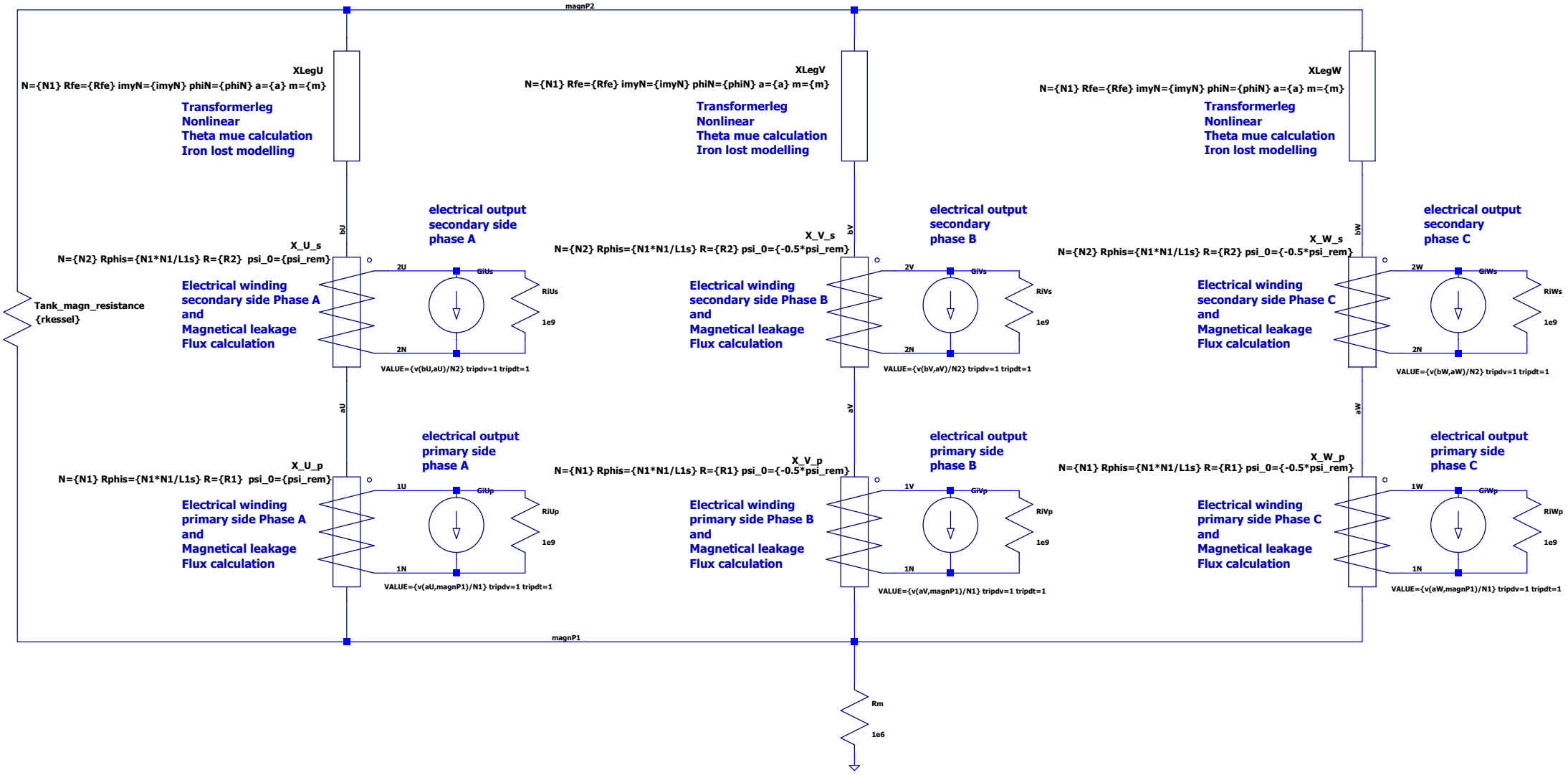Magnetic and electrical circuit